Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/29/2025
PI Name:	Patla, Bijunath Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	303-497-4082
Organization Name:	National Institute of Standards and Technology		
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PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80305	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	04/01/2021	**End Date:**	09/29/2028
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Grant/Contract number updated per D. Griffin/NASA-HQ (Ed., 1/22/25). NOTE: End date changed to 09/29/2028 per D. Griffin/NASA-HQ (Ed., 1/20/25).		
Key Personnel Changes/Previous PI:	New Principal Investigator as of April 1, 2021 is Dr. Bijunath R. Patla; previous PI was Dr. Chris Oates.		
COI Name (Institution):			
Grant/Contract No.:	80HQTR21T0082 ; NNH12AT81I		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado (PI=Oates)" with new PI, Dr. Bijunath R. Patla. To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to: 1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10^-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.		

Task Description:

2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources.

As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following:

• Global time and frequency transfer with uncertainty as low as 10^-17.

• Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10^-17 or better.

• Test of gravitation redshift at 2 x 10^-6 uncertainty or better.

• Search for variation in fundamental constants, such as fine structure constant at level of 10^-18 per year or better.

All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list):

• World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10^-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10^-16 or better.

• Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 9 x 10^-19.

• Two ytterbium lattice optical clocks with uncertainties better than 2 x 10^-18.

• Strontium lattice optical clocks with uncertainties of 2 x 10^-18 (at JILA). A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.

• A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons with demonstrated instability below 10^-16.

• A world-class time scale comprising eight hydrogen masers and a collection of Cs clocks.

NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications.

In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment.

We propose the following revised program plan outline:

Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction.

Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure.

Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time.

Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Gap Year (2017): Perform relativistic calculations to prevent unanticipated shifts from contaminating ACES data. Install updated MWL hardware into new NIST time and frequency transfer system. Prepare updated data analysis system in anticipation of ACES launch. $150k for partial support of staff time.

Gap 6 Mo. (2018): Develop and install precision time links between the MWL hardware site and NIST clocks. $80k for partial support of staff time.

Gap 12 Mo. (2018-2019): Assuming successful launch in 2020, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Develop a frequency-based time scale for ACES frequency comparisons. $80k for partial support of staff time.

Gap 12 Mo. (2019-2020): Assuming successful launch in 2021, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Evaluate and improve the performance of the new frequency-based time scale optimized for ACES frequency comparisons. $80k for partial support of staff time.

Gap 12 Mo. (2020-2021): Evaluate and improve the performance of the new frequency-based time scale. Upgrade hardware infrastructure to add resiliency and robustness to the frequency-based time scale. $80k for partial support of staff time.

Gap 13 Mo. (2021-2022):Develop and integrate methods for precise orbit determination. Accurately estimate higher-order relativistic effects,estimate time and frequency transfer noise for clocks in orbit. $90k for partial support of staff time.

Gap 12 Mo (2022-2023) Assess relevant issues such as the adequacy of knowledge of the geopotential (including dynamical effects) for supporting world-wide time comparisons with advanced clocks. $80k for partial support of staff time.

Prepare for the public release of data products in collaboration with ACES/ESA and extend investigations to benefit future missions with more precise clocks for testing fundamental physics. $93k for partial support of staff time.

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major benefits and impacts of this research program can be grouped into three main areas: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such as search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.
Task Progress:	The work has focused broadly on two main tasks: 1) Design and development of continuously running frequency standards. The National Institute of Standards and Technology (NIST) continues to develop and operate new Cs fountain standards for national timekeeping and time and frequency comparisons between metrology institutes worldwide. The latest Cs fountain from NIST is NIST-F4, slated to be commissioned as a primary frequency standard. The first evaluation of this clock is now complete, and we submitted a paper describing the process and results to a peer-reviewed journal, a requirement for the fountain to assume the status of a primary frequency standard officially. NIST-F4 will have a frequency stability of 1.5E-13/sqrt(t), where t is the averaging time, and with an uncertainty of 2.2E-16 in fractional frequency. The phase noise of the local oscillator and the quantum projection noise of the fountain limits the fountain's stability. Since mid-2024, NIST has incorporated another continuously running Cs fountain, NIST-F3, into the timescale for periodically steering UTC(NIST) by supplying corrections to the timescale algorithm (AT1) and also occasionally steering the UTC(NIST) output signals – further improving the stability of the NIST timescale. The work on the continuously operating Sr-ion clock is ongoing, and several upgrades to the laser systems are being made. We plan to operate this clock with very high uptimes before the end of the year 2025. We continue advising various United States government agencies and international organizations on clocks in space and cislunar policies with our work on lunar timekeeping. We also continue to make progress with the Yb-lattice clock by further reducing the blackbody shifts to fractional frequency under 2E-20, over a factor of 50 improvements from the current state-of-the-art lattice clocks. NIST will have a total of three Yb-lattice clocks, including a portable clock, with performance in the range E-16 to E-18 in fractional stability. 2) Upgrades to infrastructure supporting clocks and measurement systems. The power supply infrastructure for supporting clocks and time transfer equipment was upgraded to add redundancy and robustness during natural disasters or other contingencies. We commissioned a natural gas generator, transfer switch, and battery backup for the clock infrastructure. The new system should help eliminate cycle slips in clocks caused by transient events due to the aging power infrastructure, and improve the overall performance of the NIST timescale. The Liebert EXM external battery can service the clock infrastructure for 10 minutes on its own. All network cables and switches have also been upgraded, allowing access to the microwave link to be installed in May or June of 2025.
Bibliography Type:	Description: (Last Updated: 02/03/2026)
Articles in Peer-reviewed Journals	Ashby N, Patla BR. "A relativistic framework to estimate clock rates on the Moon." The Astronomical Journal. 2024 Aug 12;168(3):112. https://doi.org/10.3847/1538-3881/ad643a , Aug-2024
Articles in Peer-reviewed Journals	Bothwell T, Brand W, Fasano R, Akin T, Whalen J, Grogan T, Chen YJ, Pomponio M, Nakamura T, Rauf B, Baldoni I. "Deployment of a transportable Yb optical lattice clock." Optics Letters. 2025 Jan 14;50(2):646-9. https://doi.org/10.1364/OL.543310 , Jan-2025
Articles in Peer-reviewed Journals	Patla BR. "Reduction of magnetic perturbation for SR-POEM." Classical and Quantum Gravity. 2016 Jan 27;33(4):045006. http://dx.doi.org/10.1088/0264-9381/33/4/045006 , Jan-2016
Significant Media Coverage	Fernandez E. (Ashby N and Patla BR interview). "A time standard for the Moon—thanks to general relativity. In an interview, Drs. Ashby and Patla explain the rate of clocks on the Moon and lunar GPS." Physics, 17, p.140, September 30, 2024. https://physics.aps.org/articles/v17/140 , Sep-2024

Significant Media Coverage	Richter H (Patla B interview). "Does the Moon need its own time zone? Dr. Patla interviewed for Science magazine." Science, August 14, 2024. https://www.science.org/content/article/time-has-come-set-standardized-moon-time , Aug-2024